Chapter 2
Proofs with structure

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you will learn

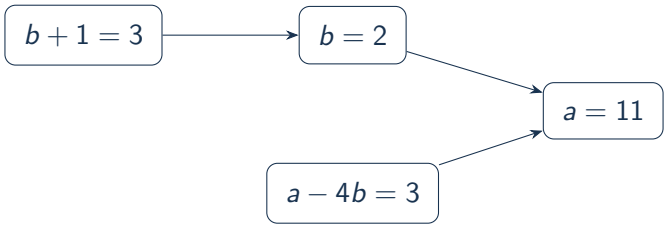
- Build and reuse an intermediate fact without losing track of the goal.
- Apply an existing lemma by checking its hypotheses.
- Use and prove statements involving $\vee$, $\wedge$, and $\exists$.
- Organize cases and witnesses in both mathematical English and Lean.

The recurring question

What information does the hypothesis provide, and what information does the goal require?

2.1 A proof can have a dependency structure

Suppose one hypothesis quickly determines b, while another relates a to b.



Prove $b = 2$ once, give it a name, and use it in the later calculation. The diagram records which facts the conclusion depends on.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.1; method of Example 2.1.1.

Worked example: an intermediate fact simplifies the calculation

Let $a, b \in \mathbb{R}$, with $a - 4b = 3$ and $b + 1 = 3$. Prove $a = 11$.

Intermediate fact

Subtracting 1 from $b + 1 = 3$ gives $b = 2$.

Now

$$\begin{aligned} a &= (a - 4b) + 4b && \text{algebra} \\ &= 3 + 4 \cdot 2 && \text{the hypothesis and intermediate fact} \\ &= 11 && \text{arithmetic.} \end{aligned}$$

The word "therefore" should identify a real dependency: this calculation uses the value of b.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.1; numerical variant of Example 2.1.1.

Lean: `have` adds a proved fact to the context

```
example {a b : Real}
    (h1 : a - 4 * b = 3) (h2 : b + 1 = 3) : a = 11 := by
  have hb : b = 2 := by addarith [h2]
  calc
    a = (a - 4 * b) + 4 * b := by ring
    _ = 3 + 4 * 2 := by rw [h1, hb]
    _ = 11 := by numbers
```

- Before `have`: only `h1` and `h2` are available.
- After it: `hb : b = 2` is also available.
- The original goal is still $a = 11$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 2.1.1; numerical variant.

Cancellation requires a reason

Example. If $t \in \mathbb{R}$, $t^2 = 4t$, and $t \geq 1$, then

$$t \cdot t = 4t, \qquad t \neq 0, \qquad \text{hence } t = 4.$$

Without $t \neq 0$, cancellation would discard the possible solution $t = 0$.

Cancelling squares is different

If $u, v \geq 0$ and $u^2 \leq v^2$, then $u \leq v$. Sign information matters: $(-3)^2 = 3^2$ does not imply $-3 = 3$.

The textbook tactic `cancel` checks the required side conditions. On paper, state them when they are doing mathematical work.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 2.1.4–2.1.5.

Practice 2.1: identify an intermediate fact

1. Let $m, n \in \mathbb{Z}$, with $m + 2 \leq 3n$ and $n \leq 4$. Prove $m \leq 10$.
2. Let $a, b \in \mathbb{R}$, with $a^2 = b^2 + 9$ and $a \geq 0$. Prove $a \geq 3$.

Write down a useful intermediate inequality before writing the final sentence. For the second problem, indicate where $a \geq 0$ is needed.

Practice 2.1: solution discussion

1. First establish $m + 2 \leq 12$:

$$m + 2 \leq 3n \leq 3 \cdot 4 = 12.$$

Subtracting 2 gives $m \leq 10$.

2. First establish the comparison of squares:

$$a^2 = b^2 + 9 \geq 9 = 3^2.$$

Since a and 3 are nonnegative, this implies $a \geq 3$. Without $a \geq 0$, the instance $a = -3, b = 0$ would contradict the conclusion.

2.2 A lemma converts one task into another

A lemma is a previously proved implication that may be reused.

Lemma	What it needs	What it gives
`ne_of_lt`	$a < b$	$a \neq b$
`ne_of_gt`	$a > b$	$a \neq b$
`le_antisymm`	$a \leq b$ and $b \leq a$	$a = b$

- `apply L` works backward: replace the goal by the hypotheses needed by L.
- `have h := L ...` works forward: store the conclusion obtained from supplied inputs.
- `exact h` closes a goal already proved by h.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.2.

Worked example: prove a stronger comparison

Let $x \in \mathbb{Q}$ and suppose $5x = 3$. To prove $x \neq 1$, it suffices to prove $x < 1$:

$$x = \frac{5x}{5} = \frac{3}{5} < 1.$$

```
example {x : Rat} (h : 5 * x = 3) : Not (x = 1) := by
  apply ne_of_lt
  calc
    x = 5 * x / 5 := by ring
    _ = 3 / 5 := by rw [h]
    _ < 1 := by numbers
```

The new goal after `apply ne_of_lt` is $x < 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.2; numerical variant of Example 2.2.1.

Worked example: squeeze a square to zero

Let $a, b \in \mathbb{R}$ and suppose $a^2 + b^2 = 0$. Prove $a^2 = 0$.

1. A square is nonnegative, so $0 \leq a^2$.
2. The other square is nonnegative, so

$$a^2 \leq a^2 + b^2 = 0.$$

3. Combining $a^2 \leq 0$ and $0 \leq a^2$ gives $a^2 = 0$.

Lean structure

`apply le_antisymm` creates two goals. Both must be proved; the first is an upper bound and the second a lower bound.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 2.2.3.

Practice 2.2: choose the lemma

1. Let $q \in \mathbb{Q}$, with $4q \leq 8$ and $2q \geq 4$. Prove $q = 2$.
2. Let $m \in \mathbb{Z}$, with $m + 3 = 7$. Prove $3m \neq 9$.

Which lemma should be applied first? What new goal or goals will it create?

Practice 2.2: solution discussion

1. Use antisymmetry. Its two required inequalities are

$$q = \frac{4q}{4} \leq \frac{8}{4} = 2, \qquad q = \frac{2q}{2} \geq \frac{4}{2} = 2.$$

Hence $q = 2$.

2. Since $m + 3 = 7$, we have $m = 4$. Therefore

$$3m = 3 \cdot 4 = 12 > 9.$$

Use `ne_of_gt` to obtain $3m \neq 9$.

2.3 “Or” has different roles in a hypothesis and a goal

$P \vee Q$ means that at least one of P, Q is true; both may be true.

Using $P \vee Q$

Prove the target assuming P. Then prove it assuming Q. Both cases are required because you do not know which holds.

Proving $P \vee Q$

It is enough to prove one alternative. Choose `left` for P or `right` for Q. The chosen alternative may depend on the current case.

Facts introduced inside one case are available only in that case.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 2.3.1–2.3.4.

Worked example: the same goal in two cases

Let $x, y \in \mathbb{R}$, and suppose $x = 2$ or $y = -2$. Prove $xy + 2x = 2y + 4$.

Case 1: $x = 2$

$$xy + 2x = 2y + 2 \cdot 2 = 2y + 4.$$

Case 2: $y = -2$

$$xy + 2x = x(-2) + 2x = 0 = 2(-2) + 4 = 2y + 4.$$

Each branch reaches the entire original goal.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.3; method of Example 2.3.1.

Worked example: create a useful disjunction

Let $x \in \mathbb{R}$ and suppose $x^2 - 5x + 6 = 0$. Prove $x = 2$ or $x = 3$.

1. Factor the known polynomial:

$$(x - 2)(x - 3) = x^2 - 5x + 6 = 0.$$

2. Use the zero-product property: $uv = 0$ implies $u = 0$ or $v = 0$.
3. If $x - 2 = 0$, then $x = 2$, establishing the left alternative.
4. If $x - 3 = 0$, then $x = 3$, establishing the right alternative.

The new disjunction is an intermediate fact; it is not an extra assumption.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.3; numerical variant of Example 2.3.4.

Lean: make the case structure visible

```
example {x : Real} (h : x ^ 2 - 5 * x + 6 = 0) :
    Or (x = 2) (x = 3) := by
  have hf : (x - 2) * (x - 3) = 0 := by
    calc
      (x - 2) * (x - 3) = x ^ 2 - 5 * x + 6 := by ring
      _ = 0 := h
  have hz := eq_zero_or_eq_zero_of_mul_eq_zero hf
  cases hz with
  | inl h2 =>
      left
      addarith [h2]
  | inr h3 =>
      right
      addarith [h3]
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.3; ASCII syntax for the case structure.

A discrete gap gives an exhaustive case split

Claim. No natural number has square 2.
For $n \in \mathbb{N}$, either $n \leq 1$ or $2 \leq n$.

- If $n \leq 1$, then $n^2 \leq 1 < 2$.
- If $2 \leq n$, then $n^2 \geq 4 > 2$.

Thus in either case $n^2 \neq 2$.

What changes for integers?

Handle negative n by considering the nonnegative number $-n$ and using $(-n)^2 = n^2$.

The split is exhaustive because no natural number lies strictly between 1 and 2.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 2.3.2 and 2.3.5.

Practice 2.3: create the right cases

1. If $x \in \mathbb{R}$ and $x^2 - 2x - 3 = 0$, prove $x = -1$ or $x = 3$.
2. For every $n \in \mathbb{N}$, prove $n^2 \neq 6$.

Explain why your cases cover all possibilities. For the first problem, say which alternative of the goal each case proves.

Practice 2.3: solution discussion

1. Since

$$(x+1)(x-3) = x^2 - 2x - 3 = 0,$$

we have $x + 1 = 0$ or $x - 3 = 0$. These give $x = -1$ and $x = 3$, respectively.

2. Split into $n \leq 2$ and $3 \leq n$.

$$n \leq 2 \implies n^2 \leq 4 < 6, \qquad 3 \leq n \implies n^2 \geq 9 > 6.$$

Every natural number lies in one of these cases.

2.4 An “and” packages two facts

$P \wedge Q$ means that both P and Q hold.

Using $P \wedge Q$

Extract both facts and use them in the same proof. For a Lean hypothesis `h`, its components are `h.1` and `h.2`.

Proving $P \wedge Q$

Prove both components. `constructor` creates two goals, or `And.intro` combines two existing proofs.

An inequality chain such as $0 < x < 1$ abbreviates $0 < x \wedge x < 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.4.

Worked example: solve for both variables

Let $x, y \in \mathbb{Q}$, with $x + y = 8$ and $x + 2y = 11$. Prove $x = 5$ and $y = 3$.

1. First find a reusable intermediate fact:

$$y = (x + 2y) - (x + y) = 11 - 8 = 3.$$

2. Use it to obtain the other component:

$$x = (x + y) - y = 8 - 3 = 5.$$

3. We have now proved both parts of the conjunction.

Establishing $y = 3$ before splitting the goal lets both parts refer to it.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.4; methods of Examples 2.4.1 and 2.4.3.

Lean: extract, derive, and repackage

```
example {x y : Rat}
    (h : And (x + y = 8) (x + 2 * y = 11)) :
    And (x = 5) (y = 3) := by
  have hy : y = 3 := by
    calc
      y = (x + 2 * y) - (x + y) := by ring
      _ = 11 - 8 := by rw [h.2, h.1]
      _ = 3 := by numbers
  have hx : x = 5 := by
    calc
      x = (x + y) - y := by ring
      _ = 8 - 3 := by rw [h.1, hy]
      _ = 5 := by numbers
  exact And.intro hx hy
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.4; new numerical example.

Practice 2.4: unpack and repack

1. Let $r, s \in \mathbb{R}$. Assume the single conjunction

$$r + s \leq 3 \ \wedge \ r - s \leq 5.$$

   Prove $r \leq 4$.

2. Let $p \in \mathbb{Z}$, with $p - 1 \geq 3$. Prove $p \geq 4 \wedge p^2 \geq 16$.

Which proof uses a conjunction? Which constructs a conjunction?

Practice 2.4: solution discussion

1. Both components of the hypothesis are available:

$$r = \frac{(r+s)+(r-s)}{2} \leq \frac{3+5}{2} = 4.$$

2. The hypothesis gives $p \geq 4$ by adding 1. In particular $p \geq 0$, so squaring preserves the comparison:

$$p^2 \geq 4^2 = 16.$$

These prove the two components of $p \geq 4 \wedge p^2 \geq 16$.

2.5 An existential statement promises a witness

$$\exists x \in D,\ P(x)$$

asserts that at least one object x in the domain D satisfies P.

Using an existential

Introduce a witness x and its promised property $P(x)$. You may use that property, but may not assume anything else about the witness.

Proving an existential

Choose a particular witness $w \in D$ and prove $P(w)$. Lean's `use w` leaves the property as the new goal.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.5.

Worked example: use a witness supplied by the hypothesis

Let $a \in \mathbb{Q}$. Suppose there is a rational number b with $a = b^2 + 2$. Prove $a > 0$.

1. Introduce such a number b, with its property $a = b^2 + 2$.
2. Every rational square is nonnegative, so

$$a = b^2 + 2 \geq 2 > 0.$$

The witness is not yours to choose

The hypothesis does not say that $b = 0$. It gives some b that satisfies the property, and the proof must work for that witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.5; numerical variant of Example 2.5.1.

Worked example: construct a point between two numbers

Let $p, q \in \mathbb{R}$ and suppose $p < q$. Prove that there exists x with $p < x < q$.

Choice of witness

Take $x = (p + q)/2$.

Check both required inequalities:

$$p = \frac{p+p}{2} < \frac{p+q}{2}, \qquad \frac{p+q}{2} < \frac{q+q}{2} = q.$$

Both steps use $p < q$ and division by the positive number 2. The domain matters: there is no integer strictly between 2 and 3.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 2.5.7.

Lean: a witness becomes a concrete verification task

```
example : Exists (fun n : Int => 9 * n = 54) := by
  use 6
  numbers

example (x : Real) : Exists (fun y : Real => y > x) := by
  use x + 1
  extra
```

In the first proof, `use 6` replaces an existence goal by $9 \cdot 6 = 54$.
In the second proof, the witness is a formula depending on the given x.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.5; methods of Examples 2.5.3–2.5.4.

Practice 2.5: propose and verify witnesses

1. Find integers m, n with $m^2 + n^2 = 65$.
2. Given any integer a, prove that some integers m, n satisfy

$$m^2 - n^2 = 2a + 1.$$

3. Given a real number t, prove that some real u satisfies $u < t$.

Write down the proposed witness before checking the property.

Practice 2.5: solution discussion

1. Take $m = 4$ and $n = 7$:

$$4^2 + 7^2 = 16 + 49 = 65.$$

2. Take $m = a + 1$ and $n = a$:

$$(a+1)^2 - a^2 = a^2 + 2a + 1 - a^2 = 2a + 1.$$

3. Take $u = t - 1$. Since $-1 < 0$, we have $t - 1 < t$.

Each proof contains two ingredients: an object of the required type and a verification of its property.

Source: Heather Macbeth, *The Mechanics of Proof*, Problem 2 follows Example 2.5.6.

Chapter 2: match the logical form to the proof action

Form	Use as a hypothesis	Prove as a goal
$P \vee Q$	Handle both cases	Establish one alternative
$P \wedge Q$	Extract both facts	Establish both components
$\exists x,\ P(x)$	Introduce witness and property	Choose and verify a witness

- `have` records intermediate work.
- A lemma application must discharge all of its required hypotheses.
- Case assumptions and witness properties have a scope; respect it.

Reading and source references

Heather Macbeth, *The Mechanics of Proof*, Chapter 2:

- 2.1 Intermediate steps
- 2.2 Invoking lemmas
- 2.3 “Or” and proof by cases
- 2.4 “And”
- 2.5 Existence proofs

https://hrmacbeth.github.io/math2001/02_Proofs_with_Structure.html

Companion code: https://github.com/hrmacbeth/math2001

These slides independently organize the chapter’s proof methods and include new numerical practice.